

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 3.1 REPRODUCTION IN ANIMALS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use this document as a MODEL for your own Final Explanation. Your final product is a comparative portfolio/poster showing animal fertilisation strategies, the human hormonal pathways, and a reasoned position paper on reproductive biotechnology. Write in your own words. Begin by linking your answer to our anchoring phenomenon: frogs near Kisumu and Lake Victoria wetlands, hens on farms, cows giving birth to calves, and Kenyan families using IVF or surrogacy. Use the correct scientific terms, draw labelled diagrams, and back every claim with evidence from the lessons. Read each section prompt, then compare your work with the exemplar. Do not copy the exemplar; use it to see what a strong answer looks like.

1. Strategies of Fertilisation: Why Frogs, Hens and Cows Differ

Compare external and internal fertilisation using the frog, hen and cow from our phenomenon. Explain why frogs release thousands of eggs while cows produce one calf. Link each strategy to the animal's environment and to how many offspring survive.

Animals reproduce by external or internal fertilisation, depending on their adaptations and environment. A frog in a pond near Kisumu releases eggs and sperm into water, where they meet outside the body. This is external fertilisation. Water stops the gametes from drying out, and it carries the sperm to the eggs. But the jelly-covered eggs have no shell, no parental care and many predators such as fish and birds, so most do not survive. Producing thousands of eggs makes sure that a few tadpoles grow into adult frogs. A hen and a cow live on land, where gametes would dry out, so they use internal fertilisation: sperm are placed inside the female's body, where fertilisation happens in moist, protected conditions. Because the embryo is protected, more survive, so fewer offspring are needed. A hen is oviparous: she lays a hard-shelled egg, and the yolk feeds the embryo outside her body. A cow is viviparous: the calf develops inside the uterus for months, fed through the placenta, and is born alive. Some animals, such as certain snakes and sharks, are ovoviviparous: the eggs hatch inside the mother's body. In short, the less protection the young get, the more gametes are produced; the more protection and parental care, the fewer offspring are needed.

2. Human Reproductive Anatomy and Gametes

Describe how the structure of human male and female reproductive systems and their gametes is suited to

Humans use internal fertilisation, like the cow. The male system makes sperm in the testes and delivers them through the sperm duct and urethra. A sperm is small and has a flagellum (tail) for swimming, many mitochondria in the middle piece to supply energy, and an acrosome at the head with enzymes to break into the egg. The female system has ovaries that release an ovum, fallopian tubes that carry it, a uterus where a baby develops, and a vagina that receives the sperm. The ovum is large because it

fertilisation. Include a labelled diagram and state where fertilisation happens.	stores nutrients (cytoplasm) to feed the early embryo, and it is surrounded by a protective layer. Unlike the frog's thousands of eggs, a woman usually releases one ovum a month, because the baby develops safely inside the body. Fertilisation happens in the fallopian tube, and the fertilised egg then moves to the uterus. My diagram labels the testis, sperm duct, urethra, ovary, fallopian tube, uterus, cervix and vagina, and I show how a sperm's structure matches its job.
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3. Hormonal Control: From Gamete Maturation to Birth

Explain how hormones control the menstrual cycle, pregnancy and birth. Name FSH, LH, oestrogen, progesterone, hCG, oxytocin and prolactin, and state what each does. Draw a flow chart showing the hormonal pathway.	Hormones work in feedback loops to coordinate human reproduction. At the start of the menstrual cycle, FSH from the pituitary gland makes a follicle in the ovary mature an ovum. The follicle makes oestrogen, which rebuilds the uterine lining and signals the pituitary to release a surge of LH. LH causes ovulation, which is the release of the ovum. The empty follicle becomes the corpus luteum, which makes progesterone to thicken and maintain the uterine lining. If there is no fertilisation, the corpus luteum breaks down, hormone levels fall and the lining is shed as menstruation. If fertilisation happens in the fallopian tube, the embryo implants in the uterus and releases hCG. hCG keeps the corpus luteum alive so progesterone stays high and the lining is not lost. Later the placenta develops. It feeds the foetus, removes wastes and also makes oestrogen and progesterone. At the end of pregnancy, oxytocin causes strong uterine contractions for childbirth, and prolactin triggers milk production. Oxytocin is also released when the baby suckles, which helps milk flow. My flow chart shows: FSH → follicle growth → oestrogen → LH surge → ovulation → progesterone → (hCG if pregnant) → placenta → oxytocin and prolactin.
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4. When Reproduction Fails: Reproductive Biotechnology

Explain how IVF, surrogacy and sperm banking help when natural reproduction fails. Correct the misconception that IVF babies grow wholly in a test tube.	Natural reproduction can fail when something blocks the process, such as blocked fallopian tubes, low sperm count, or hormonal problems that stop ovulation. Science can step in. In IVF (in-vitro fertilisation), hormones such as FSH stimulate the ovaries to mature several ova. The ova are collected and mixed with sperm in a laboratory dish, where fertilisation happens outside the body. The embryo is grown for a few days and then placed in the mother's uterus. Many people think IVF babies grow wholly in a test tube, but this is wrong. Only fertilisation and the first few days of cell division happen in the lab. The embryo must implant in a uterus and develop for about nine months, using the placenta and the same hormones, like hCG and progesterone, as in any pregnancy. In surrogacy, another woman carries the embryo when the mother cannot, for example if she has no functioning uterus. Sperm banking stores frozen sperm, for example for a man who will undergo cancer treatment that could damage his testes. These technologies are copies of natural steps, with scientists helping the parts that fail.
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5. Position Paper: Should Kenya Support Reproductive Biotechnology?

Write a short reasoned position (about 150 words) on reproductive biotechnology. State your claim, give at least two scientific reasons, include one ethical, cost or cultural consideration, and end by answering the driving question.	I believe Kenya should support access to IVF, surrogacy and sperm banking, with clear laws and counselling. First, science shows these methods work by supporting natural biology: IVF uses the same gametes, hormones and uterus as natural pregnancy, so it helps couples who have blocked tubes or low sperm count. Second, sperm banking protects fertility for patients such as cancer patients. However, we must consider cost, since IVF is expensive and most families in places like Kisumu or Kericho may not afford it. There are also ethical questions about the rights of surrogate mothers and the future child, and cultural views about family and parenthood. Laws should protect everyone and clinics should give honest information about success rates. In conclusion, animals ensure successful reproduction through different strategies: frogs produce many eggs, while hens and cows protect fewer offspring through internal fertilisation. Humans use hormones and anatomy to protect development, and when this fails, science can assist by copying and supporting these natural steps.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Comparison of animal fertilisation strategies	Accurately compares external and internal fertilisation and oviparous, viviparous and ovoviviparous development using frog, hen and cow, and clearly links gamete number to environment and offspring survival.	Correctly describes most strategies with examples, but the link between environment, protection and number of offspring is only partly explained.	Names some strategies but with errors or no examples, and does not explain why numbers of offspring differ.
Human anatomy and hormonal pathway	Gives a correct, sequenced account of FSH, LH, oestrogen, progesterone, hCG, oxytocin and prolactin, with accurate labelled diagrams and a clear flow chart showing feedback.	Describes most hormones and structures correctly, with minor omissions in the sequence or labelling.	Names a few hormones or structures but confuses their roles, or the diagram and flow chart are missing or inaccurate.
Explanation of reproductive biotechnology	Explains how IVF, surrogacy and sperm banking work and what problems they solve, and corrects the test-tube misconception with evidence about implantation and pregnancy.	Explains the technologies correctly but gives limited detail on the problems they solve or does not fully address the misconception.	Gives a vague or incorrect description of the technologies and repeats the misconception.
Position paper reasoning and Kenyan context	Presents a clear claim supported by at least two scientific reasons and one ethical, cost or cultural consideration, with relevant Kenyan examples, and answers the driving question.	States a claim with some scientific support and a brief consideration, with limited Kenyan context.	Gives an opinion with little or no scientific evidence and does not address the driving question.
Communication and use of scientific language	Portfolio or poster is well organised, uses scientific vocabulary precisely, and connects all parts back to the anchoring phenomenon.	Mostly organised and uses scientific terms correctly, with some links to the phenomenon.	Disorganised, with few scientific terms and little connection to the phenomenon.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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